A. Proofs of Theorem 4.1

Lemma A.1 (Principal eigenvector deviation under a spectral gap)**.** *We consider a symmetric matrix $H \in \mathbb{R}^{d\times d}$ with eigenvalues $\lambda_1(H) > \lambda_2(H) \geq \cdots$ and a spectral gap $\gamma := \lambda_1(H) - \lambda_2(H) > 0$. We define the perturbed matrix $H^+ := H + \Delta H$ where ΔH is symmetric, and we set the perturbation magnitude $s := \|\Delta H\|_{\mathrm{op}}$. We assume that the perturbation satisfies $s < \gamma$. We let $v_1 := v_1(H)$ and $v' := v_1(H^+)$ denote the unit principal eigenvectors of H and H^+, respectively, with the sign of v' chosen so that $v'^\top v_1 \geq 0$. Then we have*

$$\sin\angle(v', v_1) \leq \frac{s}{\gamma - s}, \qquad \|v' - v_1\| \leq \frac{2s}{\gamma - s}.$$

Proof. We let $u := v_1$ and $v := v'$. We decompose v into components parallel and orthogonal to u:

$$v = \alpha u + w, \qquad \alpha := u^\top v \geq 0, \qquad w \perp u, \qquad \|v\| = 1 \Rightarrow \alpha^2 + \|w\|^2 = 1.$$

We note that $\|w\| = \sin\angle(u, v)$.

We let $\mu_1 := \lambda_1(H^+)$ so that $(H + \Delta H)v = \mu_1 v$. Rearranging the terms, we obtain

$$(H - \mu_1 I)v = -\Delta H\, v.$$

We project onto $u^\perp$ using $P_\perp := I - uu^\top$. Since $P_\perp u = 0$ and $P_\perp v = w$, we arrive at

$$(H - \mu_1 I)w = -P_\perp \Delta H\, v.$$

Taking norms and using $\|P_\perp\| \leq 1$, we obtain

$$\|(H - \mu_1 I)w\| \leq \|\Delta H\|_{\mathrm{op}}\|v\| = s. \tag{A.1}$$

Next, we lower bound the left-hand side. By applying Weyl's inequality, we have $\mu_1 = \lambda_1(H^+) \geq \lambda_1(H) - \|\Delta H\|_{\mathrm{op}} = \lambda_1 - s$, so that

$$\mu_1 - \lambda_2 \geq (\lambda_1 - s) - \lambda_2 = \gamma - s. \tag{A.2}$$

Since $w \perp u$ lies in the invariant subspace corresponding to eigenvalues at most λ_2, we observe that the smallest eigenvalue of $(\mu_1 I - H)$ restricted to $u^\perp$ is at least $\mu_1 - \lambda_2$. Hence, we have

$$\|(\mu_1 I - H)w\| \geq (\mu_1 - \lambda_2)\|w\| \geq (\gamma - s)\|w\|. \tag{A.3}$$

Combining (A.1) and (A.3), we find $(\gamma - s)\|w\| \leq s$, i.e.,

$$\|w\| \leq \frac{s}{\gamma - s}.$$

Therefore, we conclude $\sin\angle(v', v_1) = \|w\| \leq s/(\gamma - s)$. Finally, because $v'^\top v_1 \geq 0$, we note that the angle $\theta := \angle(v', v_1) \in [0, \pi/2]$ and

$$\|v' - v_1\| = 2\sin(\theta/2) \leq 2\sin\theta = 2\|w\| \leq \frac{2s}{\gamma - s}.$$

This completes the proof. □

We consider the local geometry of a thrice-differentiable function $f : \mathbb{R}^d \to \mathbb{R}$. Let $H := \nabla^2 f(x)$ be the Hessian at x, with eigenvalues $\lambda_1(H) > \lambda_2(H) \geq \ldots$. Given an update step $\Delta x = \eta\tilde{h}$, the perturbed Hessian is $H^+ := \nabla^2 f(x + \Delta x)$, and we define the residual operator $\Delta H := H^+ - H$.

Definition A.2 (Spectral and Smoothness Parameters)**.** The following quantities characterize the stability of the system:

- **Eigengap:** $\gamma := \lambda_1(H) - \lambda_2(H) > 0$.
- **Perturbation Magnitude:** $s := \|\Delta H\|_{\mathrm{op}}$.

- **Uniform Third-derivative Bound:** A constant $M_3 > 0$ such that:

$$|\nabla^3 f(z)[a,b,c]| \le M_3 \|a\| \|b\| \|c\|, \quad \forall a,b,c \in \mathbb{R}^d.$$

Assumption 7 (Decrease Conditions)**.** *To guarantee the strict decrease of the top eigenvalue, we assume:*

- ***Non-degeneracy:*** *The perturbation is bounded by $s < \gamma$.*
- ***Descent Direction (C1):*** *$v_1^\top \Delta H v_1 \le -\eta\kappa$ for some $\kappa > 0$.*
- ***Step Size Sufficiency (C3):*** *$\eta\kappa > C_0 s$, where $C_0 := \frac{4(\|H\|_{\mathrm{op}}+s)}{\gamma - s}$ is the spectral sensitivity coefficient.*

Theorem A.3 (Strict Decrease of λ_1)**.** *Under Assumption 7, the perturbation magnitude satisfies $s \le M_3 \eta \|\tilde{h}\|$, and the leading eigenvalue of the perturbed Hessian satisfies:*

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + \frac{4(\|H\|_{\mathrm{op}} + s)}{\gamma - s}\, s. \tag{13}$$

Furthermore, the conditions (C1) and (C3) imply $\lambda_1(H + \Delta H) < \lambda_1(H)$.

Proof. We apply the fundamental theorem of calculus to the matrix-valued function $t \mapsto \nabla^2 f(x + t\Delta x)$ to write

$$\Delta H = \int_0^1 \frac{d}{dt} \nabla^2 f(w + t\Delta x)\, dt = \int_0^1 \nabla^3 f(w + t\Delta w)[\,\cdot, \cdot, \Delta w\,]\, dt. \tag{14}$$

Consequently, for any unit vectors u, v, we bound the operator norm s as follows:

$$\begin{aligned} s := \|\Delta H\|_{\mathrm{op}} &= \sup_{\|u\|=\|v\|=1} \left| u^\top \Delta H\, v \right| \\ &= \sup_{\|u\|=\|v\|=1} \left| \int_0^1 \nabla^3 f(x + t\Delta x)[u, v, \Delta x]\, dt \right| \\ &\le \sup_{\|u\|=\|v\|=1} \int_0^1 M_3 \|u\|\, \|v\|\, \|\Delta x\|\, dt \\ &= M_3 \|\Delta x\| = M_3\, \eta\, \|\tilde{h}\|. \end{aligned} \tag{15}$$

Using the Rayleigh quotient, we express the leading eigenvalue as $\lambda_1(H + \Delta H) = {v'}^\top (H + \Delta H)\, v'$. By adding and subtracting $v_1^\top (H + \Delta H)\, v_1$, we decompose this term as

$$\lambda_1(H + \Delta H) = v_1^\top (H + \Delta H)\, v_1 + \Big({v'}^\top (H + \Delta H)\, v' - v_1^\top (H + \Delta H)\, v_1 \Big). \tag{16}$$

Moreover, utilizing the eigen-equation $H v_1 = \lambda_1(H) v_1$ and the normalization $\|v_1\| = 1$, we simplify the first term:

$$v_1^\top (H + \Delta H)\, v_1 = \lambda_1(H) + v_1^\top \Delta H\, v_1. \tag{17}$$

We recall that for any symmetric matrix A and unit vectors u, v, the inequality $\left| u^\top A u - v^\top A v \right| \le 2\|A\|_{\mathrm{op}} \|u - v\|$ holds. Applying this bound to our context, we obtain

$$\begin{aligned} \left| {v'}^\top (H + \Delta H) v' - v_1^\top (H + \Delta H) v_1 \right| &\le 2\|H + \Delta H\|_{\mathrm{op}} \|v' - v_1\| \\ &\le 2(\|H\|_{\mathrm{op}} + \|\Delta H\|_{\mathrm{op}}) \|v' - v_1\| \\ &= 2(\|H\|_{\mathrm{op}} + s) \|v' - v_1\|. \end{aligned} \tag{18}$$

To estimate the eigenvector difference, we set $u := v_1$ and $v := v'$. We decompose v as

$$v = \alpha u + w, \qquad \alpha = u^\top v \ge 0, \qquad w \perp u, \qquad \alpha^2 + \|w\|^2 = 1.$$

We define $\mu_1 := \lambda_1(H + \Delta H)$. Since $(H + \Delta H)v = \mu_1 v$, we rearrange the equation to

$$(H - \mu_1 I)v = -\Delta H\, v. \tag{19}$$

We introduce $P^\perp := I - uu^\top$ as the orthogonal projector onto $u^\perp$. Applying $P^\perp$ to (19) and utilizing the properties $P^\perp u = 0$ and $P^\perp v = w$, we derive

$$(H - \mu_1 I)w = -P^\perp \Delta H\, v. \tag{20}$$

Taking the norms, we obtain

$$\|(H - \mu_1 I)w\| \le \|P^\perp\|_{\mathrm{op}} \|\Delta H\|_{\mathrm{op}} \|v\| \le s, \tag{21}$$

where we used $\|P^\perp\|_{\mathrm{op}} = 1$ and $\|v\| = 1$. Next, by invoking Weyl's inequality, we have

$$\mu_1 = \lambda_1(H + \Delta H) \ge \lambda_1(H) - \|\Delta H\|_{\mathrm{op}} = \lambda_1(H) - s, \tag{22}$$

and hence

$$\mu_1 - \lambda_2(H) \ge (\lambda_1(H) - s) - \lambda_2(H) = \gamma - s. \tag{23}$$

Since $w \perp u$, the vector w lies in the orthogonal complement of the top eigenspace of H. Therefore, we lower bound the norm as

$$\|(\mu_1 I - H)w\| \ge (\mu_1 - \lambda_2(H))\|w\| \ge (\gamma - s)\|w\|. \tag{24}$$

We note that $\|(H - \mu_1 I)w\| = \|(\mu_1 I - H)w\|$. Combining (21) and (24), we find that

$$(\gamma - s)\|w\| \le s \quad \Rightarrow \quad \|w\| \le \frac{s}{\gamma - s}.$$

Finally, since $\alpha \ge 0$ (implying the angle between u and v lies in $[0, \pi/2]$), we have $\|v - u\| \le 2\|w\|$, and thus we conclude

$$\|v' - v_1\| = \|v - u\| \le \frac{2s}{\gamma - s}. \tag{25}$$

Substituting (17) into (16) and utilizing the bounds (18) and (25), we obtain

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + 2(\|H\|_{\mathrm{op}} + s) \cdot \frac{2s}{\gamma - s}.$$

This can be rewritten as

$$\lambda_1(H + \Delta H) \le \lambda_1(H) + v_1^\top \Delta H v_1 + \underbrace{\frac{4(\|H\|_{\mathrm{op}} + s)}{\gamma - s}}_{=:C_0} s. \tag{26}$$

If condition (C1) holds, i.e., $v_1^\top \Delta H v_1 \le -\eta\kappa$, then (26) implies $\lambda_1(H + \Delta H) \le \lambda_1(H) - \eta\kappa + C_0 s$. If in addition (C3) is satisfied, i.e., $\eta\kappa > C_0 s$, then we strictly obtain

$$\lambda_1(H + \Delta H) < \lambda_1(H), \tag{27}$$

which proves the desired strict decrease. □

B. Proofs of Theorem 4.2

Definition B.1 (Algorithm Quantities)**.** We define the following quantities to characterize the convergence rate and perturbation bias:

- **Gradient Notations:** Let $g_t := \nabla f(w_t)$ be the true gradient, $h_t := \nabla f(w_t + \rho\epsilon_t)$ be the perturbed gradient, and $\hat{h}_t$ be the stochastic estimate.
- **SAM Bias:** $D_\rho := \beta^2\rho^2(1 + C_1\alpha^2)$, representing the error upper bound $\mathbb{E}\|g_t - h_t\|^2 \le D_\rho$.
- **Descent Coefficient:** $c_0 := \frac{3}{4} - 2\alpha^2$, which serves as the effective curvature scaling.

Assumption 8 (Stochastic Setup and Parameters)**.** *We adopt the following standard assumptions:*

1. ***Oracle Properties:*** *The stochastic gradient is unbiased,* $\mathbb{E}[\hat{h}_t|w_t] = h_t$*, with bounded variance* $\mathbb{E}[\|\hat{h}_t - h_t\|^2] \le \sigma^2$*.*
2. ***Perturbation Bound:*** *The noise satisfies* $\|\epsilon_t\|^2 \le 1 + C_1\alpha^2$*.*
3. ***Parameter Selection:*** *We choose* α *such that* $\alpha^2 < 3/8$ *(ensuring* $c_0 > 0$*) and set the step size as:*

$$\eta = \min\left\{\frac{1}{2\beta}, \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}, \quad \textit{where } \Delta := f(w_0) - f_*.$$

Theorem B.2 (Convergence Rate)**.** *Under Assumption 8, the X-SAM algorithm guarantees the following bound on the average squared gradient norm:*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \;\le\; c_1\left(\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1+2\alpha^2)D_\rho, \tag{28}$$

where $c_1, c_2 > 0$ *are constants depending only on the descent coefficient* c_0*.*

Proof. We assume $\alpha \in [0, 2]$ and $\|v_t\| = 1$. Consequently, for the projector $P_t := I - \alpha v_t v_t^\top$, we have the operator norm bound $\|P_t\|_{\mathrm{op}} = \max\{1, |1-\alpha|\} \le 1$. Let us define the gradient terms: $\|P_t\|_{\mathrm{op}} = \max\{1, |1-\alpha|\} \le 1$. Let

$$g_t := \nabla f(w_t), \qquad h_t := \nabla f(w_t + \rho\epsilon_t), \qquad \hat{h}_t := \nabla f(w_t + \rho\epsilon_t; \xi_t), \qquad \tilde{h}_t := P_t\hat{h}_t,$$

and the update rule $w_{t+1} = w_t - \eta\tilde{h}_t$. We assume that the stochastic gradient is unbiased with bounded variance, i.e.,

$$\mathbb{E}[\hat{h}_t \mid w_t, \epsilon_t, v_t] = h_t, \qquad \mathbb{E}\Big[\|\hat{h}_t - h_t\|^2 \mid w_t, \epsilon_t, v_t\Big] \le \sigma^2.$$

By the tower property, this implies the global statements

$$\mathbb{E}\langle u_t, \hat{h}_t - h_t\rangle = 0 \;\; (\forall\, u_t = u_t(w_t, \epsilon_t, v_t)), \qquad \mathbb{E}\|\hat{h}_t - h_t\|^2 \le \sigma^2. \tag{29}$$

Assume $\|\epsilon_t\|^2 \le 1 + C_1\alpha^2$ for some $C_1 > 0$. Since ∇f is β-Lipschitz, we have the pointwise bound

$$\|g_t - h_t\| = \big\|\nabla f(w_t) - \nabla f(w_t + \rho\epsilon_t)\big\| \le \beta\rho\|\epsilon_t\| = \beta^2\rho^2(1 + C_1\alpha^2). \tag{30}$$

By the β-smoothness of f, we have

$$f(w_{t+1}) \le f(w_t) + \langle g_t, w_{t+1} - w_t\rangle + \frac{\beta}{2}\|w_{t+1} - w_t\|^2 = f(w_t) - \eta\langle g_t, \tilde{h}_t\rangle + \frac{\beta\eta^2}{2}\|\tilde{h}_t\|^2.$$

Taking the total expectation yields

$$\mathbb{E}f(w_{t+1}) \le \mathbb{E}f(w_t) - \eta\,\mathbb{E}\langle g_t, \tilde{h}_t\rangle + \frac{\beta\eta^2}{2}\mathbb{E}\|\tilde{h}_t\|^2. \tag{31}$$

Bounding the inner product term: Since $\tilde{h}_t = P_t\hat{h}_t$, using the tower property and the definition of P_t, we derive:

$$\begin{aligned}
\mathbb{E}\langle g_t, \tilde{h}_t\rangle &= \mathbb{E}\big[\langle g_t, P_t\hat{h}_t\rangle\big] = \mathbb{E}\Big[\mathbb{E}\big[\langle g_t, P_t\hat{h}_t\rangle \mid w_t, \epsilon_t, v_t\big]\Big] \\
&= \mathbb{E}\Big[\big\langle g_t, P_t\,\mathbb{E}[\hat{h}_t \mid w_t, \epsilon_t, v_t]\big\rangle\Big] = \mathbb{E}\langle g_t, P_t h_t\rangle \\
&= \mathbb{E}\langle g_t, h_t\rangle - \alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle]\,.
\end{aligned} \tag{32}$$

Since $\|P_t\|_{\mathrm{op}} \le 1$, we have $\|\tilde{h}_t\| = \|P_t\hat{h}_t\| \le \|\hat{h}_t\|$ pointwise. Consequently, this yields the inequality:

$$\mathbb{E}\|\tilde{h}_t\|^2 \le \mathbb{E}\|\hat{h}_t\|^2. \tag{33}$$

We decompose the stochastic gradient as $\hat{h}_t = h_t + (\hat{h}_t - h_t)$ and apply (29) to derive:

$$\begin{aligned}
\mathbb{E}\|\hat{h}_t\|^2 &= \mathbb{E}\langle \hat{h}_t, \hat{h}_t\rangle \\
&= \mathbb{E}\Big\langle h_t + (\hat{h}_t - h_t),\, h_t + (\hat{h}_t - h_t)\Big\rangle \\
&= \mathbb{E}\Big[\|h_t\|^2 + 2\langle h_t, \hat{h}_t - h_t\rangle + \|\hat{h}_t - h_t\|^2\Big] \\
&= \mathbb{E}\|h_t\|^2 + 2\,\mathbb{E}\langle h_t, \hat{h}_t - h_t\rangle + \mathbb{E}\|\hat{h}_t - h_t\|^2 \\
&= \mathbb{E}\|h_t\|^2 + \mathbb{E}\|\hat{h}_t - h_t\|^2 \\
&\le \mathbb{E}\|h_t\|^2 + \sigma^2. \qquad (34)
\end{aligned}$$

Combining (33) and (34), we deduce that

$$\mathbb{E}\|\tilde{h}_t\|^2 \le \mathbb{E}\|h_t\|^2 + \sigma^2. \qquad (35)$$

Substituting (32) and (35) into (31), we arrive at

$$\mathbb{E}f(w_{t+1}) \le \mathbb{E}f(w_t) - \eta\,\mathbb{E}\langle g_t, h_t\rangle + \eta\alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle] + \frac{\beta\eta^2}{2}\mathbb{E}\|h_t\|^2 + \frac{\beta\eta^2}{2}\sigma^2. \qquad (36)$$

We utilize the polarization identity $2\langle a, b\rangle = \|a\|^2 + \|b\|^2 - \|a-b\|^2$ (pointwise) and take the expectation to write

$$-\eta\,\mathbb{E}\langle g_t, h_t\rangle = -\frac{\eta}{2}\mathbb{E}\|g_t\|^2 - \frac{\eta}{2}\mathbb{E}\|h_t\|^2 + \frac{\eta}{2}\mathbb{E}\|g_t - h_t\|^2. \qquad (37)$$

For any $\delta \in (0,1)$, we apply Young's inequality using the fact that $|v_t| = 1$ to obtain the pointwise bound:

$$\begin{aligned}
\alpha\langle h_t, v_t\rangle\langle g_t, v_t\rangle &= \Big(\alpha\langle h_t, v_t\rangle\Big)\Big(\langle g_t, v_t\rangle\Big) \\
&\le \frac{\delta}{2}\,\langle g_t, v_t\rangle^2 + \frac{1}{2\delta}\Big(\alpha\langle h_t, v_t\rangle\Big)^2 \qquad (\forall\,\delta > 0) \qquad (38) \\
&= \frac{\delta}{2}\,\langle g_t, v_t\rangle^2 + \frac{\alpha^2}{2\delta}\,\langle h_t, v_t\rangle^2 \\
&\le \frac{\delta}{2}\,\|g_t\|^2 + \frac{\alpha^2}{2\delta}\,\|h_t\|^2, \qquad (39)
\end{aligned}$$

Hence, we have

$$\eta\alpha\,\mathbb{E}[\langle h_t, v_t\rangle\langle g_t, v_t\rangle] \le \frac{\eta\delta}{2}\mathbb{E}\|g_t\|^2 + \frac{\eta\alpha^2}{2\delta}\mathbb{E}\|h_t\|^2. \qquad (40)$$

Substituting (37) and (40) into (36), we obtain

$$\begin{aligned}
\mathbb{E}f(w_{t+1}) \le\ & \mathbb{E}f(w_t) - \frac{\eta}{2}(1-\delta)\mathbb{E}\|g_t\|^2 + \frac{\eta}{2}\mathbb{E}\|g_t - h_t\|^2 \\
&+ \Big(-\frac{\eta}{2} + \frac{\eta\alpha^2}{2\delta} + \frac{\beta\eta^2}{2}\Big)\mathbb{E}\|h_t\|^2 + \frac{\beta\eta^2}{2}\sigma^2. \qquad (41)
\end{aligned}$$

From the triangle inequality $\|h_t\| \le \|g_t\| + \|h_t - g_t\|$, we have

$$\|h_t\|^2 \le 2\|g_t\|^2 + 2\|h_t - g_t\|^2. \qquad (42)$$

Taking the expectation and using (30) (which implies $\|h_t - g_t\|^2 = \|g_t - h_t\|^2 \le D_\rho$ point-wise), we obtain

$$\mathbb{E}\|h_t\|^2 \le 2\,\mathbb{E}\|g_t\|^2 + 2D_\rho, \qquad \mathbb{E}\|g_t - h_t\|^2 \le D_\rho. \qquad (43)$$

We fix $\delta = \frac{1}{2}$, $\beta\eta \le \frac{1}{2}$, and $\alpha \le \sqrt{\frac{3}{8}}$, so that we satisfy the descent condition

$$\frac{\alpha^2}{\delta} + \beta\eta \le \frac{3-\delta}{2} \qquad (44)$$

Applying (43), we obtain

$$\begin{aligned}\mathbb{E}f(w_{t+1}) &\le \mathbb{E}f(w_t) - \frac{\eta}{2}(1-\delta)\mathbb{E}\|g_t\|^2 + \frac{\eta}{2}D_\rho + \Big(-\frac{\eta}{2} + \frac{\eta\alpha^2}{2\delta} + \frac{\beta\eta^2}{2}\Big)\Big(2\,\mathbb{E}\|g_t\|^2 + 2D_\rho\Big) + \frac{\beta\eta^2}{2}\sigma^2 \\ &\le \mathbb{E}f(w_t) - \eta\Big(\frac{3-\delta}{2} - \frac{\alpha^2}{\delta} - \beta\eta\Big)\mathbb{E}\|g_t\|^2 + \eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2}{2}\sigma^2 \qquad (45)\end{aligned}$$

We define

$$c := \frac{3-\delta}{2} - \frac{\alpha^2}{\delta} - \beta\eta.$$

We fix $\delta = \frac{1}{2}$ and assume $\beta\eta \le \frac{1}{2}$ and $\alpha^2 < \frac{3}{8}$. Then for this c we have

$$c = \frac{5}{4} - 2\alpha^2 - \beta\eta \ge \frac{5}{4} - 2\alpha^2 - \frac{1}{2} = \frac{3}{4} - 2\alpha^2 =: c_0 > 0,$$

which implies $1/c \le 1/c_0$.

From (45) we directly obtain

$$\eta c\,\mathbb{E}\|g_t\|^2 \le \mathbb{E}f(w_t) - \mathbb{E}f(w_{t+1}) + \eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2}{2}\sigma^2, \qquad (46)$$

$$\begin{aligned}\eta c\sum_{t=0}^{T-1}\mathbb{E}\|g_t\|^2 &\le \sum_{t=0}^{T-1}\big(\mathbb{E}f(w_t) - \mathbb{E}f(w_{t+1})\big) + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2 \\ &= \mathbb{E}\big[f(w_0) - f(w_T)\big] + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2 \\ &\le \Delta + T\eta\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta^2 T}{2}\sigma^2,\end{aligned}$$

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{\Delta}{c\,\eta T} + \frac{1}{c}\Big(\frac{1}{2} + \frac{\alpha^2}{\delta} + \beta\eta\Big)D_\rho + \frac{\beta\eta}{2c}\sigma^2 \qquad (47)$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2} + 2\alpha^2 + \beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \qquad (48)$$

Using $\beta\eta \le \frac{1}{2}$, we further obtain

$$\Big(\frac{1}{2} + 2\alpha^2 + \beta\eta\Big)D_\rho \le (1 + 2\alpha^2)\,D_\rho.$$

Moreover, by the definition of η, we have

$$\frac{\Delta}{\eta T} \le 2\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}, \qquad \beta\eta\sigma^2 \le \sqrt{\frac{\beta\Delta\sigma^2}{T}}.$$

We choose

$$\eta = \min\left\{\frac{1}{2\beta},\ \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}.$$

Plugging these bounds into (48), we obtain that there exist constants $c_1, c_2 > 0$ (depending only on c_0) such that

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le c_1\left(\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1 + 2\alpha^2)\,D_\rho, \qquad D_\rho = \beta^2\rho^2(1 + C_1\alpha^2). \qquad (49)$$

In particular, we note that when $\sigma^2 > 0$ the stochastic term scales as $O(1/\sqrt{T})$, and when $\sigma = 0$ the bound reduces to $O(1/T)$ up to the perturbation bias term proportional to D_ρ.

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{\Delta}{c\,\eta T} + \frac{1}{c}\Big(\frac{1}{2}+\frac{\alpha^2}{\delta}+\beta\eta\Big)D_\rho + \frac{\beta\eta}{2c}\sigma^2 \tag{50}$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2}+\frac{\alpha^2}{\delta}+\beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2$$

$$\le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}\Big(\frac{1}{2}+2\alpha^2+\beta\eta\Big)D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \tag{51}$$

Using $\beta\eta \le \frac{1}{2}$, we further obtain

$$\Big(\tfrac{1}{2}+2\alpha^2+\beta\eta\Big)D_\rho \le (1+2\alpha^2)\,D_\rho. \tag{52}$$

Substituting (52) into (48), we obtain

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{1}{c_0}\cdot\frac{\Delta}{\eta T} + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho + \frac{1}{c_0}\cdot\frac{\beta\eta}{2}\sigma^2. \tag{53}$$

We choose the step size $\eta = \min\left\{\frac{1}{2\beta},\ \sqrt{\frac{\Delta}{\beta\sigma^2 T}}\right\}$. Then $\beta\eta \le \frac{1}{2}$ and, moreover, $\frac{\Delta}{\eta T} \le 2\frac{\beta\Delta}{T} + \sqrt{\frac{\beta\Delta\sigma^2}{T}}, \beta\eta\sigma^2 \le \sqrt{\frac{\beta\Delta\sigma^2}{T}}$. We obtain

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla f(w_t)\|^2 \le \frac{1}{c_0}\left(2\frac{\beta\Delta}{T}+\sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho + \frac{1}{2c_0}\sqrt{\frac{\beta\Delta\sigma^2}{T}}$$

$$= \frac{2}{c_0}\frac{\beta\Delta}{T} + \frac{3}{2c_0}\sqrt{\frac{\beta\Delta\sigma^2}{T}} + \frac{1}{c_0}(1+2\alpha^2)\,D_\rho \tag{54}$$

$$\le c_1\left(\frac{\beta\Delta}{T}+\sqrt{\frac{\beta\Delta\sigma^2}{T}}\right) + c_2(1+2\alpha^2)\,D_\rho.$$

□

Theorem B.3 (First-order change of the leading eigenvalue)**.** *We assume that $f:\mathbb{R}^d\to\mathbb{R}$ is $\mathcal{C}^4$ in a neighborhood of w_t, and we define $H(w)=\nabla^2 f(w)$ and $H_t := H(w_t)$. We assume the existence of $\rho>0$ such that the third and fourth derivatives are locally bounded: $M_3 := \sup_{\|u-w_t\|\le\rho}\|\nabla^3 f(u)\|_{\mathrm{op}} < \infty$, $M_4 := \sup_{\|u-w_t\|\le\rho}\|\nabla^4 f(u)\|_{\mathrm{op}} < \infty$. We further assume that H_t possesses a simple largest eigenvalue $\lambda_1(w_t)$ with unit eigenvector $v_1(w_t)$, and we define the spectral gap as $\gamma := \lambda_1(w_t)-\lambda_2(w_t) > 0$. We consider the update $\Delta w := w_{t+1}-w_t = -\eta g_t$ with $\eta>0$ chosen sufficiently small to ensure that $\|H(w_t+\Delta w)-H(w_t)\|_{\mathrm{op}} < \gamma/2$. We additionally assume that $\|(I-v_1v_1^\top)\nabla\lambda_1(w_t)\| \le \varepsilon$, and$v_1(w_t)^\top g_t = 0$. Under these assumptions, we establish that the leading eigenvalue increment $\Delta\lambda_1 := \lambda_1(w_{t+1})-\lambda_1(w_t)$ satisfies*

$$|\Delta\lambda_1| \le \eta\varepsilon\|g_t\| \;+\; O\big(\eta^2\|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2\|g_t\|^2}{\gamma}\Big),$$

where we note that the hidden constants depend only on M_3, M_4 and on the universal spectral constant in the perturbation remainder.

Proof. We let $H(w)=\nabla^2 f(w)$ and $H_t := H(w_t)$. We define the updates as

$$\Delta w := w_{t+1}-w_t = -\eta g_t, \qquad \Delta H := H(w_t+\Delta w)-H(w_t).$$

By invoking the *analytic perturbation theorem* (Kato, 1966), we assert that there exists a neighborhood of H_t (in particular, whenever $\|\Delta H\|_{\mathrm{op}} < \gamma/2$) on which $\lambda_1(\cdot)$ is differentiable and

$$D\lambda_1(H_t)[E] = v_1^\top E\, v_1.$$

Hence, the first-order expansion yields

$$\Delta\lambda_1 := \lambda_1(w_{t+1}) - \lambda_1(w_t) = v_1^\top(\Delta H)\, v_1 + r_{\text{spec}}, \tag{55}$$

with the gap-controlled remainder satisfying

$$|r_{\text{spec}}| \le C_{\text{spec}} \frac{\|\Delta H\|_{\text{op}}^2}{\gamma}. \tag{56}$$

We let $\mathcal{B}$ be a ball around w_t containing the segment $\{w_t + s\Delta w : s \in [0,1]\}$, and we define the Lipschitz constants

$$L_0 := \sup_{w\in\mathcal{B}} \|DH(w)\|_{\text{op}}, \qquad L_1 := \sup_{w\in\mathcal{B}} \|D^2H(w)\|_{\text{op}}.$$

(In particular, if $\|\nabla^3 f(w)\|$ is bounded on $\mathcal{B}$ by M_3, then $L_0 \le M_3$. If moreover $\|\nabla^4 f(w)\|$ is bounded on $\mathcal{B}$ by M_4, then $L_1 \le M_4$.) Consequently, we bound the perturbation norm as

$$\|\Delta H\|_{\text{op}} = \|H(w_t + \Delta w) - H(w_t)\|_{\text{op}} \le L_0\|\Delta w\|. \tag{57}$$

We define $\phi(s) := H(w_t + s\Delta w)$. Then, noting that $\phi'(s) = DH(w_t + s\Delta w)[\Delta w]$, we write

$$\Delta H = \int_0^1 DH(w_t + s\Delta w)[\Delta w]\, ds. \tag{58}$$

Adding and subtracting $DH(w_t)[\Delta w]$, we obtain

$$\Delta H = DH(w_t)[\Delta w] + R_H, \qquad R_H := \int_0^1 \big(DH(w_t + s\Delta w) - DH(w_t)\big)[\Delta w]\, ds. \tag{59}$$

In coordinates, we express the differential as

$$DH(w_t)[\Delta w] = \sum_{i=1}^{d} \frac{\partial H}{\partial w_i}(w_t)\, \Delta w_i =: \nabla_w H[w_t] \cdot \Delta w. \tag{60}$$

Since D^2H is bounded by L_1 on $\mathcal{B}$, the map DH is Lipschitz on $\mathcal{B}$ with constant L_1, i.e.,

$$\|DH(u) - DH(w)\|_{\text{op}} \le L_1\|u - w\| \qquad \forall\, u, w \in \mathcal{B}.$$

Therefore, for all $s \in [0,1]$, we have

$$\|DH(w_t + s\Delta w) - DH(w_t)\|_{\text{op}} \le L_1\|s\Delta w\| = sL_1\|\Delta w\|.$$

Substituting this bound into the definition of R_H, we derive

$$\begin{aligned}
\|R_H\|_{\text{op}} &\le \int_0^1 \Big\| \big(DH(w_t + s\Delta w) - DH(w_t)\big)[\Delta w] \Big\|_{\text{op}}\, ds \\
&\le \int_0^1 \|DH(w_t + s\Delta w) - DH(w_t)\|_{\text{op}}\, \|\Delta w\|\, ds \\
&\le \int_0^1 sL_1\|\Delta w\|^2\, ds = \tfrac{1}{2}L_1\|\Delta w\|^2.
\end{aligned} \tag{61}$$

Combining (55) and (59), we arrive at

$$\Delta\lambda_1 = v_1^\top\big(DH(w_t)[\Delta w]\big)v_1 \;+\; v_1^\top R_H v_1 \;+\; r_{\text{spec}}. \tag{62}$$

Using $\|v_1\| = 1$ and (61), we find

$$|v_1^\top R_H v_1| \le \|R_H\|_{\text{op}} \le \tfrac{1}{2}L_1\|\Delta w\|^2.$$

Moreover, by applying (56) and (57), we obtain

$$|r_{\text{spec}}| \le C_{\text{spec}} \frac{\|\Delta H\|_{\text{op}}^2}{\gamma} \le C_{\text{spec}} \frac{L_0^2}{\gamma} \|\Delta w\|^2.$$

Expanding $DH(w_t)[\Delta w]$ using (60) gives

$$v_1^\top \big(DH(w_t)[\Delta w]\big) v_1 = \sum_{i=1}^{d} \Delta w_i \, v_1^\top \frac{\partial H}{\partial w_i}(w_t) v_1.$$

We let $H_t := H(w_t)$ and suppose the top eigenpair (λ_1, v_1) satisfies $H_t v_1 = \lambda_1 v_1, \qquad \|v_1\|_2 = 1$. We differentiate both sides with respect to the coordinate w_i (evaluated at w_t). Using the product rule, we obtain

$$\frac{\partial H}{\partial w_i}(w_t)\, v_1 \;+\; H_t \frac{\partial v_1}{\partial w_i} = \frac{\partial \lambda_1}{\partial w_i}(w_t)\, v_1 \;+\; \lambda_1 \frac{\partial v_1}{\partial w_i}. \tag{63}$$

Multiplying by $v_1^\top$ from the left, we obtain

$$v_1^\top \frac{\partial H}{\partial w_i}(w_t)\, v_1 \;+\; v_1^\top H_t \frac{\partial v_1}{\partial w_i} = \frac{\partial \lambda_1}{\partial w_i}(w_t)\, v_1^\top v_1 \;+\; \lambda_1\, v_1^\top \frac{\partial v_1}{\partial w_i}. \tag{64}$$

Since H_t is symmetric, we note that

$$v_1^\top H_t \frac{\partial v_1}{\partial w_i} = \lambda_1\, v_1^\top \frac{\partial v_1}{\partial w_i}, \tag{65}$$

so these terms cancel on both sides. Therefore, we derive

$$\frac{\partial \lambda_1}{\partial w_i}(w_t) = v_1^\top \frac{\partial H}{\partial w_i}(w_t)\, v_1, \qquad i = 1, \ldots, d. \tag{66}$$

and hence

$$v_1^\top \big(DH(w_t)[\Delta w]\big) v_1 = \sum_{i=1}^{d} \Delta w_i \frac{\partial \lambda_1}{\partial w_i}(w_t) = \langle \nabla \lambda_1(w_t),\, \Delta w \rangle. \tag{67}$$

Substituting (67) into (62) and using $\Delta w = -\eta g_t$, we obtain

$$\Delta \lambda_1 = -\eta \langle \nabla \lambda_1(w_t),\, g_t \rangle \;+\; v_1^\top R_H v_1 \;+\; r_{\text{spec}}.$$

Using the bounds above and the relation $\|\Delta w\| = \eta \|g_t\|$, we have

$$\Delta \lambda_1 = -\eta \langle \nabla \lambda_1(w_t),\, g_t \rangle \;+\; O\big(\eta^2 \|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2 \|g_t\|^2}{\gamma}\Big),$$

where the hidden constants depend only on the local smoothness constants L_0, L_1 and on C_{spec}.

We decompose the gradient into the component orthogonal to v_1 and the component parallel to v_1:

$$\nabla \lambda_1(w_t) = (I - v_1 v_1^\top) \nabla \lambda_1(w_t) + v_1 v_1^\top \nabla \lambda_1(w_t). \tag{68}$$

Taking the inner product with Δw, we obtain

$$\langle \nabla \lambda_1(w_t), \Delta w \rangle = \langle (I - v_1 v_1^\top) \nabla \lambda_1(w_t), \Delta w \rangle + \langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \rangle \tag{69}$$

By applying the Cauchy–Schwarz inequality, we have

$$\big|\langle (I - v_1 v_1^\top) \nabla \lambda_1(w_t), \Delta w \rangle\big| \le \|(I - v_1 v_1^\top) \nabla \lambda_1(w_t)\| \, \|\Delta w\| \le \varepsilon \|\Delta w\|. \tag{70}$$

Moreover, using the identity $v_1 v_1^\top x = (v_1^\top x)\, v_1$ for any vector x, we observe that

$$v_1 v_1^\top \nabla \lambda_1(w_t) = \big(v_1^\top \nabla \lambda_1(w_t)\big)\, v_1, \tag{71}$$

$$\big\langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \big\rangle = \big(v_1^\top \nabla \lambda_1(w_t)\big)\, \big(v_1^\top \Delta w\big), \tag{72}$$

$$\big|\big\langle v_1 v_1^\top \nabla \lambda_1(w_t), \Delta w \big\rangle\big| \le \big|v_1^\top \nabla \lambda_1(w_t)\big| \, \big|v_1^\top \Delta w\big|. \tag{73}$$

Therefore, in the orthogonal case where $v_1^\top g_t = 0$, we deduce

$$\big|\eta\langle \nabla\lambda_1(w_t), g_t\rangle\big| = \big|\langle \nabla\lambda_1(w_t), \Delta w\rangle\big| \le \varepsilon\|\Delta w\| = \eta\varepsilon\|g_t\|.$$

Finally, we obtain the single-step eigenvalue change bound

$$|\Delta\lambda_1| \le \eta\varepsilon\|g_t\| \;+\; O\big(\eta^2\|g_t\|^2\big) \;+\; O\Big(\frac{\eta^2\|g_t\|^2}{\gamma}\Big),$$

where we note that the hidden constants depend only on the local smoothness constants L_0, L_1 and on C_{spec}. Hence, we conclude that when ε is sufficiently small, the first-order contribution is suppressed and λ_1 changes only by higher-order terms in one step. □

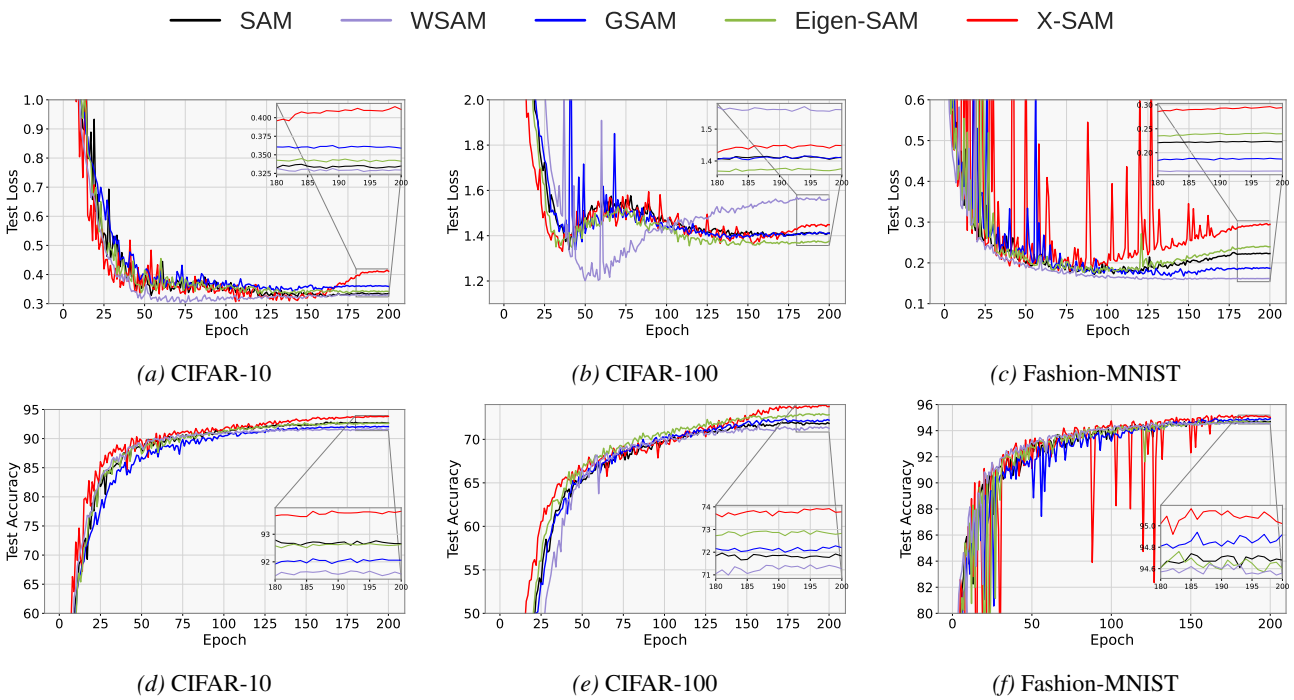
C. Additional Experimental Results

In this appendix, we present comprehensive empirical evaluations to further validate the effectiveness and robustness of the proposed X-SAM. First, we detail the evolution of test loss and accuracy across various architectures (refer to Section C.1), providing insights into the decoupling of loss minimization and generalization improvement. Then, we conduct sensitivity analyses on key hyperparameters, specifically investigating the impact of the perturbation radius ρ (explicated in Section C.2) and the gradient correction coefficient α (analyzed in Section C.3). Lastly, we perform a Hessian spectrum investigation (outlined in Section C.4) to visualize the eigenvalue distribution, confirming that X-SAM successfully locates flatter minima by suppressing the dominant eigenvalue.

C.1. Evaluation on Test Loss and Accuracy

We evaluated the test accuracy and test loss of various methods using the ResNet-50, WideResNet, and AlexNet architectures. The experimental results are presented in Figure 7. As indicated by the accuracy curves, X-SAM consistently outperforms SAM, GSAM, and other baseline methods across all tested architectures and datasets, demonstrating superior generalization capabilities. Notably, we observed an intriguing phenomenon during our experiments: although X-SAM achieves the highest test accuracy, its corresponding test loss is not always the lowest; in some instances, it is marginally higher than that of standard SAM or other variants.

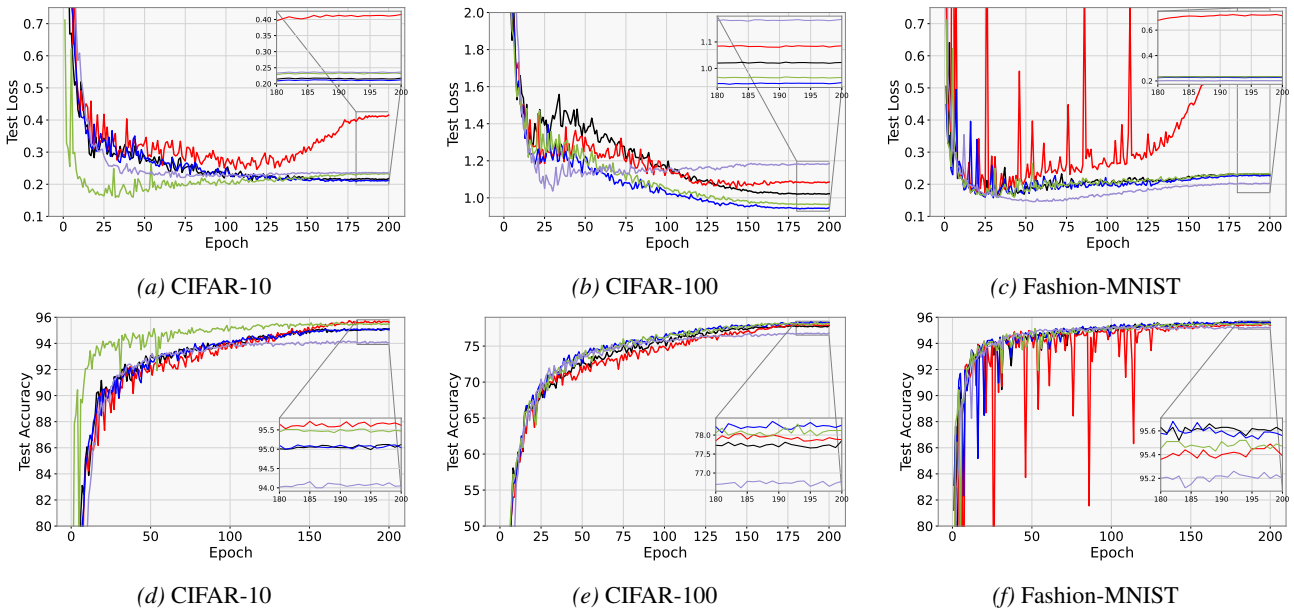
This phenomenon of "high accuracy accompanied by relatively higher loss" suggests that X-SAM does not merely pursue the minimization of cross-entropy loss (i.e., reducing the probability confidence error of predictions). Instead, it prioritizes the optimization of decision boundary robustness. A lower test loss often implies that the model is extremely "confident" (potentially over-confident) in its predictions, which can be an indicator of overfitting. Conversely, by explicitly penalizing high-curvature regions, X-SAM guides the optimization process into flatter minima. In such regions, the model exhibits more cautious predictions on misclassified samples, thereby avoiding extreme probability distributions on outliers. Consequently, the slightly elevated test loss actually reflects X-SAM's enhanced generalization ability and noise robustness, achieved while maintaining correct classification.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

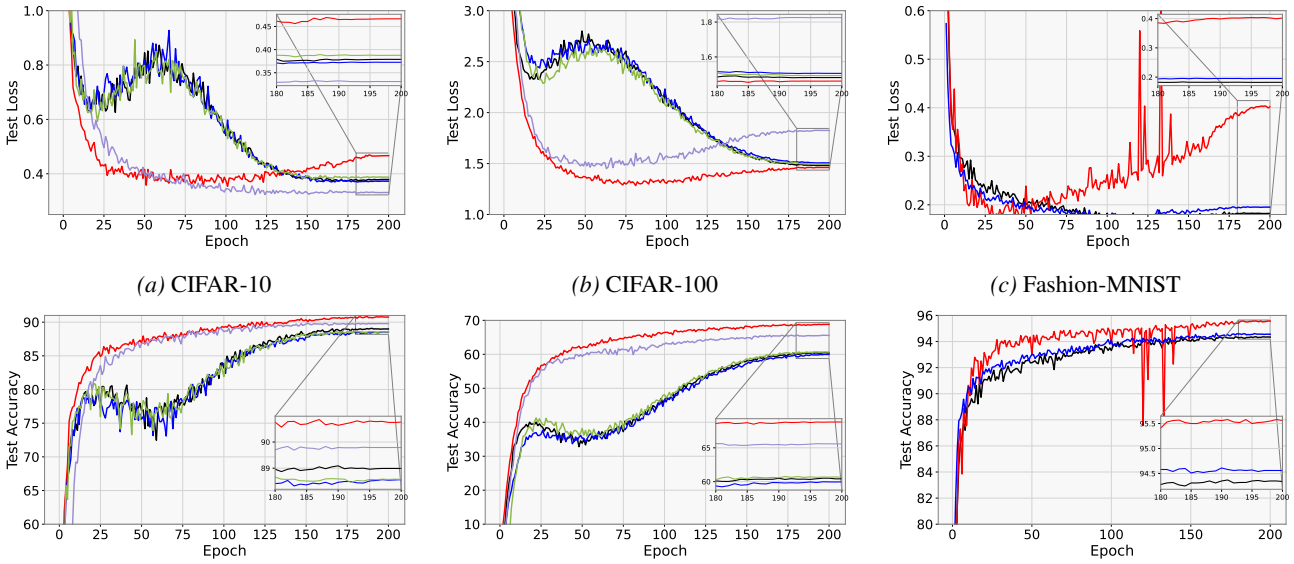
Figure 5. Learning curves for training ResNet-50 on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 5a, 5b and 5c: Test loss vs. Epochs; Figs 5d, 5e and 5f: Test accuracy vs. Epochs.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

Figure 6. Learning curves for training WideResNet on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 6a, 6b and 6c: Test loss vs. Epochs; Figs 6d, 6e and 6f: Test accuracy vs. Epochs.



(a) CIFAR-10 *(b)* CIFAR-100 *(c)* Fashion-MNIST

(d) CIFAR-10 *(e)* CIFAR-100 *(f)* Fashion-MNIST

Figure 7. Learning curves for training AlexNet on CIFAR-10, CIFAR-100, and Fashion-MNIST. Figs 7a, 7b and 7c: Test loss vs. Epochs; Figs 7d, 7e and 7f: Test accuracy vs. Epochs.

C.2. Robustness to Perturbation Radius

To investigate the effect of X-SAM on SAM's sensitivity to the perturbation radius ρ, we conducted experiments on the CIFAR-10 dataset using the ResNet-18 architecture. Specifically, we compared the test performance of SAM and X-SAM under different perturbation radii $\rho \in \{0.05, 0.1, 0.15, 0.2\}$, repeating each setting three times to report the averaged test accuracy. The figure 8 shows that as the perturbation radius increases, SAM's test accuracy initially decreases and then slightly recovers, whereas X-SAM maintains a generally stable performance. Even at $\rho = 0.1$, X-SAM preserves consistent generalization. These findings indicate that X-SAM alleviates SAM's sensitivity to the perturbation radius, enabling more stable training dynamics and consistent generalization across a wide range of ρ values.

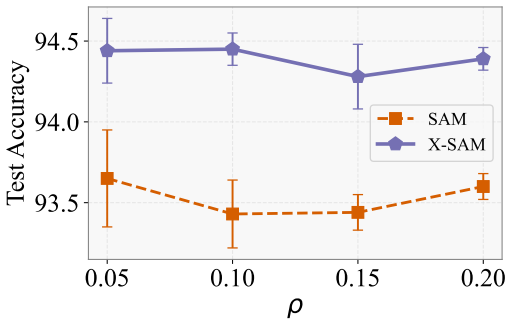


Figure 8. Comparison of test accuracy under different values of ρ on CIFAR-10 with the ResNet-18 model

C.3. Hyperparameter Sensitivity Investigation of alpha

We explore how the hyperparameter α influences the performance of X-SAM by running experiments on the CIFAR-100 dataset with the ResNet-18 architecture. Different values of α are evaluated, as illustrated in Figure 9. The results indicate that when $\alpha = 0.2$, the model achieves the highest test accuracy among all tested settings, showing a 1.07% improvement over standard SAM. This suggests that selecting an appropriate α can significantly enhance performance. Moreover, the test accuracy remains relatively stable when α varies within the range $[0.1, 0.4]$, indicating that α is a fairly robust hyperparameter.

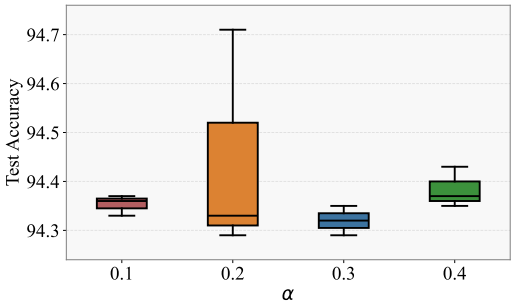


Figure 9. Test accuracy of ResNet-18 when training on CIFAR-10 under different values of α for X-SAM.

C.4. Hessian Spectrum Investigation

We analyze the impact of training with SAM and X-SAM on the Hessian spectrum of the model. The experiment is conducted on a ResNet-18 model trained on the CIFAR-100 dataset, where the top k eigenvalues of the Hessian matrix $\nabla^2 f(w)$ are computed at the second epoch and visualized. The results, as shown in Figure 10, show that compared with SAM, X-SAM consistently produces a smaller largest eigenvalue and a smaller trace, with a higher concentration of eigenvalues near zero. This indicates that X-SAM guides the optimization towards flatter regions of the loss surface, confirming the effectiveness of the proposed gradient correction mechanism and explaining why Eigen-SAM achieves better generalization performance than SAM.

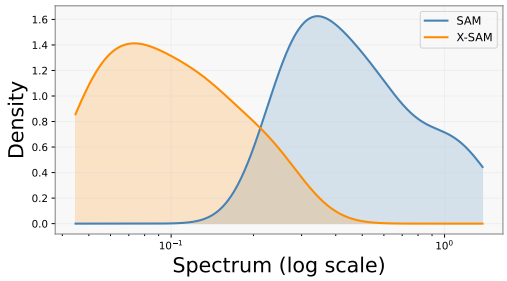


Figure 10. Hessian spectrum at the end of training ResNet-18 on CIFAR-100.